# Media Bias ISM: Master Methods & Results Table

**Shared notation.** $i$ firm, $t$ day/week. $\alpha_i$ firm FE, $\delta_t$ time (day/month) FE. $\text{Post}_t = \mathbb{1}[t \geq 2020\text{-}01\text{-}01]$. $s_a = p_a^+ - p_a^-$ (NewsMTSC article stance); $\text{bias}_{it} = |\mathcal{A}_{it}|^{-1} \sum_a s_a$; $\Delta\text{bias}_{it}$ first difference (used in VAR); $\text{vol}_{it} = |\mathcal{A}_{it}|$ (firm-day articles); $\text{return}_{it} = \ln P_{it} - \ln P_{i,t-1}$ (weekly: summed). Inference: HC1 (panel regressions); HAC/Newey–West (VAR).
**Keys.** Verdict: **sig.** ($p < 0.05$); marg. ($0.05 \leq p < 0.10$); n.s. ($p \geq 0.10$); n/a descriptive/no test. Fit: within-$R^2$ for regressions; VAR uses a Granger $F$-test, so $R^2$ is **n/a**. *[check]* construction/sanity check (shown first in each RQ). [legacy] superseded sample, retained for reference. Corpora (RQ4): **All-art:** all scored articles; **Rel.:** ensemble relevance-filtered (121,177 / 695,731).

| ID | Method / spec | Equation | Sample | Headline result | Fit ($R^2$) | Verd. |
|---|---|---|---|---|---|---|
| **RQ1: Did the *tone* of media coverage of S&P 500 firms shift after Jan 2020?** Outcome: daily firm-day stance $\text{bias}_{it}$. *Construction: $s_a = p^+ - p^-$ (NewsMTSC), high-confidence outputs, averaged per firm-day.* | | | | | | |
| RQ1.C1 | *[check]* Pre-period mean equality | Tolerance rule on firm pre-period stance means | 26 firms; $\tau$=0 | Means *unequal* (range 0.66, std 0.137) ⇒ SCDiD fallback considered for clean identification. | n/a | n/a |
| RQ1.C2 | *[check]* Joint $F$-test on day FE | $H_0 : \delta_1 = \cdots = \delta_T = 0$ | df (4,017, 59,181) | $F$=1.135, $p=1.1 \times 10^{-8}$; $F_{\text{crit}}$=1.038. Day FE jointly significant ⇒ justifies two-way FE (RQ1.1). | n/a | **sig.** |
| RQ1.C3 | *[check]* MLR / Gauss–Markov diagnostics | Breusch–Pagan; Jarque–Bera; VIF; Durbin–Watson on RQ1.1 residuals | $n$=63,200 | BP $p<10^{-42}$ (heterosk.); JB $p\approx0$ (fat tails, kurt 4.25); VIF $\approx$ 1.0; DW = 1.673. ⇒ HC1 justified. | n/a | n/a |
| RQ1.1 | Two-way FE (firm + *day*) | $\text{bias}_{it} = \alpha_i + \delta_t + \beta\, \text{Post}_t + \gamma\, \text{vol}_{it} + \varepsilon_{it}$ | $n$=63,200; 26 firms; 4,018 days; $\tau$=0 | $\beta_{\text{Post}} = +0.0141$; HC1 SE 0.0136, $p$=0.30; day-clustered SE 0.0134, $p$=0.29. vol $-0.00097^{***}$. Post insignificant once daily common shocks absorbed. | 0.0017 | n.s. |
| RQ1.1b | Two-way FE, log-volume control | $\text{bias}_{it} = \alpha_i + \delta_t + \beta\, \text{Post}_t + \gamma\, \ln(1+\text{vol}_{it}) + \varepsilon_{it}$ | $n$=63,200; $G$=4,018 | $\beta_{\text{Post}} = +0.0146$ (SE 0.0134), $p$=0.28; ln-vol $-0.0161^{***}$. Robustness to volume scaling. | 0.0014 | n.s. |
| RQ1.4 | Monthly event study on stance | $\text{bias}_{it} = \alpha_i + \sum_{k\neq-1} \beta_k \mathbb{1}[t \in \text{mo}_k] + \gamma\, \text{vol}_{it} + \varepsilon_{it}$ | $n$=23,439; $k \in [-24, +24]$; ref $k=-1$ | Pre-trend $\approx$ 0. $\beta_0$=+0.031 ($p$=0.09); $\beta_{+12}$=+0.064 ($p<10^{-3}$); persists +0.05 to +0.10 to $k$=+24. | n/a | **sig.** |
| RQ1.L | [legacy] Confidence-threshold sweep, one-way firm FE | $\text{bias}_{it} = \alpha_i + \beta\, \text{Post}_t + \gamma_1 \text{vol}_{it} + \gamma_2 \text{vix}_t + \varepsilon_{it}$ (no day FE) | $\tau$=0/0.6/0.9; $n$=49,136 / 46,681 / 31,575; 22 firms | $\beta_{\text{Post}}$=+0.0208 / +0.0232 / +0.0258 (per $\tau$); all $p<10^{-6}$. Significant on the earlier one-way-FE / 22-firm sample; superseded by RQ1.1 (effect vanishes once day FE added). | 0.0018–0.0026 | **sig.** |
| **RQ2: Did firm-level *returns* shift after Jan 2020 beyond market co-movement?** Outcome: daily $\text{return}_{it}$. *Construction: $\text{return}_{it} = \ln P_{it} - \ln P_{i,t-1}$ from `stock_prices` close; market controls S&P 500 return + VIX.* | | | | | | |
| RQ2.C1 | *[check]* Pre-period mean equality | Tolerance rule on firm pre-period return means | 24 firms | Means *equal* (range 0.0019, std 0.0004) ⇒ parallel-trends plausible; standard DiD valid, SCDiD not triggered. | n/a | n/a |
| RQ2.1 | Baseline firm FE on returns | $\text{return}_{it} = \alpha_i + \beta\, \text{Post}_t + \gamma_1 \text{sp500\_ret}_t + \gamma_2 \text{vix}_t + \varepsilon_{it}$ | $n$=62,667; 26 firms; 2015-12 to 2025-11 | $\beta_{\text{Post}} = -0.000344$ (SE 0.000172), $p$=0.045 ($\approx -3.4$ bps/day); S&P $\beta=1.128^{***}$; VIX n.s. | n/r | marg. |
| RQ2.2 | Two-way FE (firm + *month*) | $\text{return}_{it} = \alpha_i + \delta_t + \beta\, \text{Post}_t + \gamma_1 \text{sp500\_ret}_t + \gamma_2 \text{vix}_t + \varepsilon_{it}$ | $n$=62,667 | $\beta_{\text{Post}} = -0.0018$ (SE 0.0142), $p$=0.90; not significant once month FE included. S&P ret absorbed by month FE ($\beta\approx0$). | 0.0000[1] | n.s. |

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| ID | Method / spec | Equation | Sample | Headline result | Fit ($R^2$) | Verd. |
|---|---|---|---|---|---|---|
| RQ2.3 | [legacy] Monthly event study on returns | $\text{return}_{it} = \alpha_i + \sum_{k \neq -1} \beta_k \mathbb{1}[t \in \text{mo}_k] + \gamma_1 \text{sp500\_ret}_t + \gamma_2 \text{vix}_t + \varepsilon_{it}$ | $n$=62,667; ref Dec 2019 | Pre-trend flat. $\beta_0$ (Jan 2020) = −0.00145, $p$=0.044; post bins mostly n.s. *(no result file in* `results/rq2/`*; verify before citing.)* | n/a | marg. |
| **RQ3: Did the bias→return link *strengthen* post-2020?** Interpretation of the pre/post Granger split (no new estimation; draws on RQ4). *Split: firm-specific Bai–Perron single break in the bias series (see RQ3.C1), not a common Jan 2020 date.* | | | | | | |
| RQ3.C1 | *[check]* Structural breaks (Bai–Perron) | Single largest mean shift per firm in the bias series | 16 firms | Breaks span *2018–2024* (only V ≈ Sep 2020). "Pre-break" ≠ "pre-COVID" ⇒ calendar reading needs a fixed-date robustness check. | n/a | n/a |
| RQ3.1 | Pre vs. post-break Granger shift (from RQ4) | Compare Bias→Returns Granger $p$ across the firm-specific break | panel + firm; both corpora | **All-art:** panel pre $p$=0.245 → post 0.617 (no change); firm WFC pre 0.345 → post 0.013**. **Rel.:** panel pre 0.009*** → post 0.672; firms MS/SBUX significant *pre*-break, none post. Effect is **pre**-break, not strengthening after 2020. | n/a | marg. |
| **RQ4: Does media bias *predict* returns at all?** Weekly VAR($p$=3) + Granger causality (HAC SE), 15 firms. Both corpora reported. *Construction: ensemble relevance filter (both corpora); weekly ISO aggregation + gap imputation; Δbias (bias I(1)→differenced); weekly log returns winsorized ±5SD/firm (excl. AT&T); Helmert FE removal.* | | | | | | |
| RQ4.C1 | *[check]* Stationarity (ADF + KPSS) | ADF $H_0$: unit root; KPSS $H_0$: stationary | 16 firms, weekly | Returns I(0) all 16. Bias levels I(1)/"conflicting" 14/16 ⇒ VAR uses Δbias. | n/a | n/a |
| RQ4.C2 | *[check]* VAR lag selection & residual diagnostics | BIC lag order; stability roots; Portmanteau; Jarque–Bera; ARCH | 15 firms | Modal BIC lag $p$=3; all stable. Portmanteau fails 11/15; JB fails all; ARCH 10/15 ⇒ HAC/Newey–West SE throughout. | n/a | n/a |
| RQ4.1 | Panel VAR Granger: Bias → Returns | Pooled VAR on $[\Delta\text{bias}_{it}, \text{return}_{it}]$; Helmert FE; +VIX +S&P ret + article count | pooled firm-weeks; 15 firms | **All-art:** $p$=0.976 / 0.245 / 0.617 (full/pre/post); $N$=7,493. **Rel.:** $p$=0.114 / **0.009*** / 0.672; $N$=6,976. Significant pre-break only under filtering. | n/a ($F$-test) | **sig.** |
| RQ4.2 | Panel VAR Granger: Returns → Bias | (reverse direction of RQ4.1) | 15 firms | **All-art:** $p$=0.200 / 0.801 / 0.187. **Rel.:** $p$=0.706 / 0.359 / 0.131. Reverse channel null in both corpora. | n/a ($F$-test) | n.s. |
| RQ4.3 | Firm Granger: Bias → Returns (sig. firms) | Per-firm bivariate VAR, HAC SE; full / pre / post-break | ≥ 100 wks/firm | **All-art:** WFC full 0.023**, post 0.013**; MS full 0.058*. **Rel.:** SBUX full 0.024**, pre 0.014**; MS pre **0.001*****; no firm sig. post-break. | n/a ($F$-test) | **sig.** |
| RQ4.4 | Firm Granger: Returns → Bias (sig. firms) | Per-firm bivariate VAR, HAC SE | ≥ 100 wks/firm | **All-art:** GS full 0.012** / post 0.007***; MS post 0.016**; BA post 0.014**. **Rel.:** GS full 0.032** / pre 0.010***; GM full 0.079*. Price→coverage for financials. | n/a ($F$-test) | **sig.** |
| RQ4.5 | Firm Granger: remaining firms | Per-firm bivariate VAR, HAC SE | both corpora | ABNB, AMZN, T, BAC, INTC, MCD, MSFT, UBER, V: no significant Granger in either direction (either corpus). WFC dropped (coverage); WMT excluded (VAR spec). | n/a ($F$-test) | n.s. |

**Cross-cutting reading.** Static pre/post dummies are significant on the one-way firm-FE design (RQ1.L) but weaken or vanish once daily common shocks are absorbed (RQ1.1, RQ2.2); a single Post dummy averages five heterogeneous post-shock years into one coefficient. The monthly event study (RQ1.4) shows the stance shift is gradual, rising to $\beta_{+12}$=+0.064 by year 1, while returns show a one-month hit at the event date. The weekly VAR (RQ4) is run on two corpora. On all scored articles, media bias does not predict returns at the panel level ($p$=0.976). On relevance-filtered articles the effect is significant in

[1] RQ2.2 within-$R^2$ ≈ 0 because month fixed effects absorb the S&P 500 return (the dominant regressor); the model's explanatory power is loaded onto the FE, not the within variation.

the pre-break period only (panel $p$=0.009; firms MS, SBUX) and null post-break ($p$=0.672), which is the reverse of a post-2020 strengthening (RQ3). Because the structural breaks are firm-specific and span 2018 to 2024 (RQ3.C1), "pre-break" is not the same as "pre-COVID," so the calendar reading needs a fixed-date robustness check. Reverse causality (returns$\rightarrow$bias) is significant only for financials (GS, MS), consistent with price moves preceding coverage.